

AI Multimodal Emotion Monitor & Therapist

Solution Report

Project	AI Multimodal Emotion Monitor & Therapist
Team	Group 1 (5 Members)
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Abstract

This report presents the design, architecture, machine-learning pipeline, and deployment strategy for an AI-powered real-time emotion monitoring and therapy system. The system integrates facial expression recognition (DeepFace / MediaPipe), voice arousal analysis (Whisper / librosa), and wearable biometric signals (EDA/PPG) into a unified multimodal emotion profile. A Large Language Model (LLM) acts as the cognitive orchestration core, providing empathetic, context-aware therapeutic recommendations through a live Streamlit dashboard. The solution is containerised with Docker and designed for edge deployment on portable compute hardware.

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1. Introduction and Problem Statement

Traditional behavioural analysis and stress monitoring heavily rely on self-reporting or manual observation to gauge emotional distress. This subjective feedback loop is slow and often inaccurate, making it difficult to detect critical emotional shifts in real-time or provide timely interventions. Existing commercial solutions address, at most, a single modality (typically facial expressions or voice), leaving significant gaps in accuracy when any one channel is noisy or unavailable.

This project introduces the **AI Multimodal Emotion Monitor and Therapist** -- a novel affective-computing system designed for high-accuracy distress detection. The primary objective is to monitor and analyse emotional signals in real-time using a combination of physiological, vocal, and visual data, and to feed those signals to an LLM that responds empathetically as a virtual therapist.

1.1 Scope and Objectives

- * Detect seven discrete emotions (Happy, Sad, Angry, Fear, Surprise, Disgust, Neutral) and three fatigue states (Drowsiness, Yawning, Head-Nodding) from continuous video.
- * Analyse microphone audio for speaking patterns and arousal level.
- * Acquire wearable biometric data (EDA/GSR and PPG/heart-rate) for physiological grounding.
- * Fuse all three modalities into a unified real-time Distress Level score.
- * Generate empathetic, context-aware therapeutic text using an LLM (Groq / LLaMA 3).
- * Deploy the full system on portable edge hardware (Jetson / NUC) or a standard PC.

2. System Architecture

The system is designed as a continuous analysis pipeline connecting the user, the sensors, the ML classification engines, and a central LLM orchestration hub.

2.1 High-Level Flow

```
[ User / Subject ]
+---> Handback Sensors ---> Biometric Module ---+
+---> Integrated Mic ---> Voice Module ---+---> Fusion Engine --> LLM --> Dashboard
+---> Kit Camera ---> Video Module ---+
```

Figure 1 - Multimodal system architecture.

2.2 Component Descriptions

Component	Technology	Output
Video Module	DeepFace + MediaPipe	Emotion label + confidence
Voice Module	Whisper + librosa	Arousal level + transcription
Biometric Module	PySerial + CWT scalograms	EDA/PPG distress score
Fusion Engine	LangChain agent	Unified distress vector
LLM Therapist	Groq / LLaMA 3 / Gemini	Therapeutic text response
Dashboard	Streamlit	Live UI + timeline charts

2.3 Step-by-Step Monitoring Journey

- * **Calibration:** Baseline physiological and vocal data are recorded for neutral-state reference.
- * **Continuous Data Acquisition:** Sensors stream 1D physiological data, audio, and video in parallel.
- * **Feature Processing:** 1D biometrics to 2D CWT Scalograms; voice to Mel-Spectrograms; video to facial vectors.
- * **Multimodal Classification:** Each modality model outputs an emotion probability vector.
- * **LLM Feedback:** Fusion Engine combines vectors; LLM interprets context and generates recommendations.
- * **Logging and Intervention:** Events are timestamped and stored; alerts displayed on the dashboard.

3. Machine Learning Pipeline

3.1 Facial Emotion Recognition (Video)

DeepFace wraps CNN backbones (VGG-Face, FaceNet, OpenFace, DeepID) and outputs probabilities across seven emotion categories. MediaPipe Face Mesh provides 468 facial landmarks used to compute Eye Aspect Ratio (EAR) for drowsiness detection and mouth openness for yawn detection.

3.2 Voice Arousal Analysis

Audio is captured with PyAudio at 16 kHz. faster-whisper (CTranslate2 backend) converts speech to text for semantic context. librosa extracts MFCCs, spectral centroid, and zero-crossing rate to produce an arousal score on 2-second sliding windows with 50% overlap.

3.3 Biometric Physiological Processing

EDA and PPG are sampled via wearable sensor over USB serial. Each 10-second window is converted to a 2D Scalogram image using Continuous Wavelet Transform (CWT). Pre-trained MobileNet classifiers validated on DEAP and WESAD predict valence and arousal -- achieving up to 99% accuracy, far superior to raw 1D time-series classifiers.

3.4 Why 2D Vision Models for 1D Signals?

Approach	Accuracy (DEAP)	Notes
1D CNN (raw signal)	60-70%	Misses time-frequency interactions
LSTM / GRU	65-75%	Slow convergence, vanishing gradient
2D CNN + Spectrogram (STFT)	80-88%	Captures frequency patterns
2D CNN + Scalogram (CWT)	90-99%	Best: multi-resolution time-frequency

Table 1 - Accuracy comparison for biometric signal classification.

4. LLM and RAG Integration

4.1 LLM Therapist Core

An LLM acts as the cognitive core -- interpreting the fused Distress Level and producing empathetic recommendations. Supported backends:

- * **Groq (LLaMA 3 / Mixtral)** -- primary cloud backend; ultra-low latency.
- * **Local LLaMA / Gemma** -- fully offline via llama-cpp-python; no API key required.
- * **Google Gemini** -- high-capability alternative via google-genai SDK.
- * **Mock mode** -- pre-written responses for demo/testing without any API key.

4.2 Retrieval-Augmented Generation (RAG)

The core/rag/ subsystem grounds the LLM in curated mental-health knowledge:

- * Vector DB (Chroma / FAISS) stores embedded psychology books, therapy guidelines, and news articles.
- * SentenceTransformers encodes user queries for semantic similarity retrieval.
- * LangChain orchestrates the retrieval, prompt injection, and LLM call chain.
- * NewsAPI integration fetches relevant real-time articles on demand.

4.3 Prompt Engineering

Emotion vectors are serialised and injected into the LLM system prompt at each inference step. Example:

```
System: You are an empathetic AI therapist. State: Emotion=Anxious(0.78) |  
Arousal=High | EDA Distress=0.82 Provide a calming, solution-focused response in 2-3  
sentences.
```

5. Real-Time Intervention and Alerting

Three intervention modes are supported:

- * **Predictive Alerting:** Rising stress trends are flagged before visible to a human monitor.
- * **Automated Logging:** Timestamped natural-language event descriptions explain distress spikes.
- * **Intervention Recommendations:** Personalised suggestions pushed to the dashboard.
- * **TTS Output:** Recommendations spoken aloud via pyttsx3 for hands-free monitoring.

5.1 Dashboard Layout

- * Left panel: Live webcam feed with emotion label overlay and EAR/yawn indicators.
- * Centre panel: Real-time emotion timeline chart (last 60 seconds).
- * Right panel: AI Therapist chat interface with session history.
- * Status bar: Distress Level gauge and active modality indicators.

6. Datasets, Libraries and SDKs

6.1 Benchmark Datasets

Dataset	Modality	Size	Labels
FER-2013	Image	35K frames	7 discrete emotions
AffectNet	Image	1M+ images	8 emotions + valence/arousal
RAVDESS	Audio	7.4K clips	8 emotions
IEMOCAP	Audio + Video	12 hours	9 emotions + dimensional
DEAP	EEG + EDA/PPG	32 subjects	Valence / Arousal / Dominance
WESAD	Multi-physiological	15 subjects	Stress / Neutral / Amusement
MELD	Video + Text	13K utterances	7 emotions (Friends TV)

6.2 Core Python Libraries

Library	Role
DeepFace	Facial emotion recognition engine
MediaPipe 0.10.9	Face mesh, landmark, EAR computation
faster-whisper	Offline speech-to-text
librosa	Audio feature extraction (MFCC, arousal)
LangChain (+ community)	LLM orchestration and RAG pipeline
FAISS / Chroma	Vector similarity search for RAG
sentence-transformers	Text embedding for RAG
transformers (HuggingFace)	Pre-trained emotion/NLP models
OpenCV	Video frame capture and processing
Streamlit	Live dashboard UI
llama-cpp-python	Run local LLaMA on CPU/GPU
google-gemai	Google Gemini API integration
pyttsx3	Text-to-speech output
FastAPI + Uvicorn	REST / WebSocket microservices

PySerial	Serial communication with biometric hardware
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6.3 APIs and SDKs

SDK / API	Role	Auth
Groq SDK (langchain-groq)	Fast LLM inference	GROQ_API_KEY
HuggingFace Hub	Download pre-trained models	Optional token
Google GenAI SDK	Gemini model access	Google API Key
NewsAPI	Real-time news for RAG	NEWS_API_KEY
OpenAI Whisper (offline)	Speech recognition	None (local)

7. Hardware Deployment

7.1 Target Platforms

- * **Laptop / Desktop PC** -- development and demo environment.
- * **NVIDIA Jetson Nano / Xavier** -- GPU-accelerated edge inference.
- * **Intel NUC (Maven Kit)** -- portable compute unit for field deployment.
- * **Raspberry Pi 5** -- ultra-low-power sensor gateway option.

7.2 Docker Deployment

```
# Build and start docker-compose up -d # Open dashboard http://localhost:8501
```

The image is built in three layers: Dockerfile.base (heavy Python deps), Dockerfile.app (application code), and docker-compose.yml for orchestration. A split-service compose (docker-compose-split.yml) separates ML inference from the dashboard for independent scaling.

7.3 Offline Transfer (USB / SCP)

- * Export: `docker save ... | gzip > emotion-monitor.tar.gz` (~4-5 GB).
- * Copy to Kit via USB drive or SCP.
- * Import: `docker load < emotion-monitor.tar.gz`.
- * Run with device flags: `/dev/video0, /dev/snd, /dev/ttyUSB0`.

7.4 Hardware Checklist

Device	Interface	Data Type	Module
USB / CSI Camera	V4L2 / CSI	RGB video frames	modules/video/
Microphone	ALSA / PortAudio	PCM audio 16 kHz	modules/voice/
Handback Biometric Sensor	USB Serial	EDA + PPG	modules/biometrics/
Display / HDMI Monitor	HDMI	Dashboard output	Streamlit

8. Team and Responsibilities

Member	Role	Key Deliverables	Files
Member 1	Vocal Intelligence Lead	Audio stream, Mel-Spectrogram, voice model	modules/voice/
Member 2	Biometric Engineering Lead	Sensor_reader.ino, CWT Scalogram, PPG file	hardware/, modules/biometrics/
Member 3	Visual Emotion Developer	Video capture, FER pipeline, ONNX optimisation	modules/video/
Member 4	LLM and Cognitive Architect	Fusion Engine, LLM integration, RAG setup	core/model/, core/rag/
Member 5	System Integration and Deployment	Code deployment, API, dashboard, Docker	main.py, configs/

Integration Workflow

Step	Action	Responsible
1	Parallel capture of Voice, Biometrics, Video	Members 1, 2, 3
2	Modality models generate emotion vectors	Members 1, 2, 3
3	Fusion Engine combines vectors to Distress Level	Member 4
4	LLM interprets context and generates response	Member 4
5	Dashboard displays alerts and therapist response	Member 5
6	Logs saved to Kit for session review	Member 5

9. Related Work

9.1 Facial and Video Emotion Recognition

Mollahosseini et al. (2019) introduced AffectNet -- the largest in-the-wild facial expression dataset (1M+ images) -- showing deep CNNs dramatically outperform classical ML. Li and Deng (2022) surveyed CNN, GAN, graph-network, and Vision Transformer architectures for FER, directly informing frame-level vs. temporal modelling decisions. Kosti et al. (2017) demonstrated that scene context alongside face/body analysis improves emotion prediction across 26 categories -- relevant when faces are partially occluded.

9.2 Speech Emotion Recognition

Baevski et al. (2020) introduced wav2vec 2.0, achieving top results on IEMOCAP and RAVDESS via self-supervised masking of raw audio. Hsu et al. (2021) built HuBERT, capturing tonal and rhythmic content critical for speaker-independent SER. Fan et al. (2022) confirmed SSL backbones consistently outperform Mel-Spectrogram CNN baselines with lightweight variants viable for real-time deployment.

9.3 Physiological Signal Emotion Recognition

The DEAP dataset (Koelstra et al., 2012) defines canonical EDA/PPG emotion benchmarks. WESAD (Schmidt et al., 2018) provides an ecologically valid wearable benchmark confirming multimodal physiological fusion outperforms any single channel. On et al. (2023) validated our specific pipeline -- CWT scalograms + MobileNet -- achieving 99.19% valence accuracy on DEAP.

9.4 Multimodal Fusion

MuT (Tsai et al., 2019) introduced cross-modal attention for temporally unaligned audio-visual-text fusion. TDFNet (2023) extended this with hierarchical multi-scale transformers, achieving state-of-the-art on CMU-MOSI/MOSEI. Zhu et al. (2023) surveyed 100+ papers, noting physiological signals remain underrepresented -- a gap our system addresses.

9.5 LLM-Driven Affective Systems

Picard (1997) coined Affective Computing and defined the valence-arousal-dominance representation used throughout. Zhang et al. (2023) argue LLMs provide the reasoning capacity needed to move from emotion detection to adaptive emotional response. Wang et al. (2024) demonstrated that emotion-aware prompt injection significantly improves empathic quality of LLM therapist responses -- directly validating our closed-loop design.

10. Conclusion

The AI Multimodal Emotion Monitor and Therapist provides a cohesive, high-accuracy architecture for real-time emotional decoding and empathetic intervention. By combining:

- * DeepFace + MediaPipe facial analysis for robust visual emotion and fatigue detection,
- * Whisper + librosa audio processing for voice arousal and speech understanding,
- * CWT Scalogram + vision-model biometric classification for physiological grounding,
- * LangChain + LLM orchestration with RAG-augmented knowledge for therapeutic output,
- * Docker containerisation and edge-hardware deployment for real-world portability,

-- the platform delivers objective emotional insights far superior to any single-modality or self-reporting approach. The architecture is modular: each subsystem can be independently upgraded as better models emerge. This system is positioned as a viable solution for clinical monitoring, safety-critical environments, and modern behavioural research.

11. References

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